

Dictionary:

1. Create a dictionary for a product with keys: name, price, brand. Print each key-value pair.

```
product = {  
    "name": "Laptop",  
    "price": 50000,  
    "brand": "Dell"  
}  
  
for key, value in product.items():  
    print(key, ":", value)
```

```
name : Laptop  
price : 50000  
brand : Dell
```

2. Add a new key 'in_stock' with value True. Print the updated dictionary.

```
product = {  
    "name": "Laptop",  
    "price": 50000,  
    "brand": "Dell"  
}  
  
product["in_stock"] = True  
  
print(product)
```

```
{'name': 'Laptop', 'price': 50000, 'brand': 'Dell', 'in_stock': True}
```

3. Update the price to a new value and print the dictionary again.

```
product = {  
    "name": "Laptop",  
    "price": 50000,  
    "brand": "Dell",  
    "in_stock": True  
}  
  
product["price"] = 55000  
  
print(product)
```

```
{'name': 'Laptop', 'price': 55000, 'brand': 'Dell', 'in_stock': True}
```

4. Ask the user to enter a student's name and marks in 3 subjects. Store in a dictionary and print.

```
name = input("Enter student name: ")

maths = int(input("Enter Maths marks: "))
science = int(input("Enter Science marks: "))
english = int(input("Enter English marks: "))

student = {
    "Name": name,
    "Maths": maths,
    "Science": science,
    "English": english
}

print(student)
```

Enter student name: akila
Enter Maths marks: 85
Enter Science marks: 90
Enter English marks: 95
{'Name': 'akila', 'Maths': 85, 'Science': 90, 'English': 95}

5. Create a dictionary of 5 countries and their capitals. Print only the country names using keys().

```
countries = {
    "India": "New Delhi",
    "Japan": "Tokyo",
    "USA": "Washington DC",
    "Australia": "Canberra",
    "France": "Paris"
}

print("Countries:")

for country in countries.keys():
    print(country)
```

Countries:
India
Japan
USA
Australia
France

6. Traverse a dictionary of 4 students and their scores. Print each name and score using items().

```
students = {
```

```
"Akila": 92,  
"Ravi": 85,  
"Priya": 90,  
"Kiran": 88  
}  
  
for name, score in students.items():  
    print(name, ":", score)
```

```
Akila : 92  
Ravi : 85  
Priya : 90  
Kiran : 88
```

Sets:

1. Create a set of 5 numbers with duplicates. Print it and observe that duplicates are removed.

```
numbers = {1, 2, 2, 3, 4, 4, 5}  
  
print(numbers)
```

```
{1, 2, 3, 4, 5}
```

2. Add the number 100 to the set and print it.

```
numbers = {1, 2, 3, 4, 5}  
  
numbers.add(100)  
  
print(numbers)
```

```
{1, 2, 3, 4, 5, 100}
```

3. Try remove() on a value that is not in the set. Then try discard(). Write down what happens.

```
numbers = {1, 2, 3, 4, 5}  
  
numbers.remove(10)
```

```
-----  
KeyError                                Traceback (most recent call last)  
/tmp/ipykernel_8780/64804868.py in <cell line: 0>()  
      1 numbers = {1, 2, 3, 4, 5}  
      2  
----> 3 numbers.remove(10)  
  
KeyError: 10
```

Next steps: [Explain error](#)

KeyError: 10

Using discard():

```
numbers = {1, 2, 3, 4, 5}  
  
numbers.discard(10)  
  
print(numbers)  
  
{1, 2, 3, 4, 5}
```

4. Create set A of subjects you like and set B of subjects your friend likes. Find subjects both of you like.

```
A = {"Maths", "Science", "English", "Computer"}  
B = {"Science", "Computer", "History", "Telugu"}  
  
common_subjects = A.intersection(B)  
  
print("Subjects both like:", common_subjects)  
  
Subjects both like: {'Science', 'Computer'}
```

5. Using the same sets, find all subjects that either of you likes.

```
A = {"Maths", "Science", "English", "Computer"}  
B = {"Science", "Computer", "History", "Telugu"}  
  
all_subjects = A.union(B)  
  
print("All subjects:", all_subjects)  
  
All subjects: {'Maths', 'History', 'English', 'Computer', 'Telugu', 'Science'}
```

6. Convert this list to a set to remove duplicates: [1, 2, 2, 3, 4, 4, 5]. Print the result.

```
numbers = [1, 2, 2, 3, 4, 4, 5]

unique_numbers = set(numbers)

print(unique_numbers)

{1, 2, 3, 4, 5}
```

Functions:

1. Write a function square(n) that returns the square of n. Test it with 3, 5, and 10.

```
def square(n):
    return n * n

print(square(3))
print(square(5))
print(square(10))

9
25
100
```

2. Write a function is_positive(n) that returns True if n > 0, and False otherwise.

```
def is_positive(n):
    return n > 0

print(is_positive(10))
print(is_positive(-5))
print(is_positive(0))

True
False
False
```

3. Write a function that takes a list of numbers and returns their average.

```
def average(numbers):
    return sum(numbers) / len(numbers)

nums = [10, 20, 30, 40, 50]

print("Average:", average(nums))

Average: 30.0
```

4. Write a function `count_vowels(word)` that counts and returns the number of vowels in a word.

```
def count_vowels(word):  
    vowels = "aeiouAEIOU"  
    count = 0  
  
    for char in word:  
        if char in vowels:  
            count += 1  
  
    return count  
  
print(count_vowels("Rahul"))  
print(count_vowels("Python"))
```

```
2  
1
```

Write a function that takes a name and prints it reversed.

```
def reverse_name(name):  
    print(name[::-1])  
  
reverse_name("Rahul")
```

```
luhaR
```

6. Write `celsius_to_fahrenheit(c)` to convert temperature.

```
def celsius_to_fahrenheit(c):  
    return (c * 9/5) + 32  
  
print(celsius_to_fahrenheit(25))  
print(celsius_to_fahrenheit(0))
```

```
77.0  
32.0
```

7. Write a function that takes a dictionary of student names and marks, and returns the topper's name.

```
def topper(students):  
    return max(students, key=students.get)  
  
students = {  
    "Akila": 92,  
    "Ravi": 85,  
    "Priya": 95,  
    "Kiran": 88
```

```
}  
  
print("Topper:", topper(students))
```

Topper: Priya